

# AI-Powered Phishing Detection System

## Problem Statement and Literature Survey

### 1. Problem Statement

Phishing attacks are a major cybersecurity threat in which attackers use deceptive URLs and websites to trick users into revealing credentials, financial information and other sensitive data. Traditional blacklist, signature and rule-based approaches are limited against newly created or previously unseen phishing URLs. Machine-learning and deep-learning techniques can learn patterns from URL and website characteristics and automate classification.

Nevertheless, existing AI-based phishing detectors face important challenges including dataset dependency, weak generalisation to unseen URL distributions, false positives, computational cost and vulnerability to adversarially modified phishing websites. The five selected IEEE Access studies address these issues from complementary perspectives: hybrid machine learning, hybrid deep learning, adversarial robustness, proactive detection of parked domains, and cross-dataset validation.

Therefore, the proposed project aims to develop an AI-powered phishing detection system that extracts relevant URL features, trains and compares suitable AI models, evaluates them using multiple metrics and emphasises generalisation to unseen data. The system should provide a clear phishing/legitimate decision together with a useful risk assessment.

### 2. Literature Survey

#### 1. Phishing Detection System Through Hybrid Machine Learning Based on URL

**Author(s):** Abdul Karim; Mobeen Shahroz; Khabib Mustofa; Samir Brahim Belhaouari; S. Ramana Kumar Joga (2023)

**Algorithm / Methodology:** URL-based supervised ML. Evaluates DT, Logistic Regression, Random Forest, Naive Bayes, Gradient Boosting, KNN and SVC. Proposes the LSD ensemble (LR + SVC + DT) with soft and hard voting. Uses Canopy Feature Selection, Grid Search hyperparameter optimisation and cross-fold validation.

**Advantages:** Combines complementary classifiers; systematic feature selection and tuning; compares many baselines; reports accuracy, precision, recall, F1 and specificity; suitable for structured URL features.

**Disadvantages / Limitations:** Primarily URL-feature based, so it does not directly model rich webpage/visual context. Evaluation is centered on the selected dataset rather than a strong independent cross-dataset test. The ensemble and tuning pipeline is more complex than a single classifier.

#### 2. Enhancing Phishing Detection: A Novel Hybrid Deep Learning Framework for Cybercrime Forensics

**Author(s):** Faisal S. Alsubaei; Abdulwahab Ali Almazroi; Nasir Ayub (2024)

**Algorithm / Methodology:** Hybrid deep learning. SMOTE is used for class imbalance. An Ensemble AutoEncoder with ResNet (EARN) performs feature extraction, while a ResNeXt-embedded GRU (RNT) model performs classification. Hyperparameters are optimised using the Jaya optimisation method (RNT-J).

**Advantages:** Learns nonlinear representations; combines feature extraction and sequence modelling; handles imbalance with SMOTE; includes hyperparameter optimisation; reports 98% accuracy with SMOTE and improved results over comparison models.

**Disadvantages / Limitations:** Architecture is considerably more complex and computationally demanding than conventional ML. Reported results are tied to the experimental dataset, so independent cross-dataset generalisation remains an open concern. Synthetic SMOTE samples may not perfectly reflect real-world phishing distributions.

#### 3. A Study on Adversarial Sample Resistance and Defense Mechanism for Multimodal Learning-Based Phishing Website Detection

**Author(s):** Phan The Duy; Vo Quang Minh; Bui Tan Hai Dang; Ngo Duc Hoang Son; Nguyen Huu Quyen; Van-Hau Pham (2024)

**Algorithm / Methodology:** Evaluates ML, DL, ensemble and multimodal detectors. Proposes Adversarial Website Generation (AWG), including WGAN-based adversarial sample generation and transfer-based black-box attacks, and studies defense mechanisms. Uses diverse phishing/legitimate sources including OpenPhish, PhishTank, Phishing Database and Alexa.

**Advantages:** Directly addresses adversarial robustness; tests multiple model families; uses diverse data sources; demonstrates that multimodal detection can be more resistant to adversarial manipulation; provides a practical security perspective beyond ordinary accuracy.

**Disadvantages / Limitations:** Real adversarial phishing websites are scarce, motivating generated adversarial samples. GAN/WGAN generation and multimodal experiments add substantial complexity and computation. The work targets adversarial robustness and is therefore not a complete substitute for ordinary URL-only detection.

#### 4. Machine Learning for Early Detection of Phishing URLs in Parked Domains: An Approach Applied to a Financial Institution

**Author(s):** Jaqueline D. Duarte; Pedro Chagas Junior; João Paulo Javidi da Costa; Elena J. da Costa; Laerte Peotta de Melo; Rafael Rabelo Nunes; Carlos Gabriel V. N. Soares; Thiago Erivan da Cunha Silva (2025)

**Algorithm / Methodology:** Proactive ML framework, “Phishing Hunter”, for newly registered and parked domains. Uses preprocessing, feature engineering, feature selection and model optimisation. A Light Gradient Boosting Machine (LightGBM) is the main high-performing classifier. Dataset: 211,659 URLs from real-time SSL certificate monitoring, popular-domain lists and phishing incident reports; 10-fold cross-validation.

**Advantages:** Focuses on early detection before widespread blacklisting; large real-world-oriented dataset; combines feature engineering and selection; LightGBM achieved 97.28% accuracy and 96.02% recall on the balanced dataset.

**Disadvantages / Limitations:** Scope is specifically centred on newly registered/parked domains and a financial-institution use case. Cross-dataset generalisation is not the primary evaluation design. Performance depends on timely availability and quality of domain/SSL-related features.

#### 5. XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation

**Author(s):** Milosz Misiek; Tomasz Hyla (2026)

**Algorithm / Methodology:** X-PHIDE uses XGBoost with allowlist/blocklist checks and URL feature extraction. More than 90 URL/DNS/security/reputation/behaviour features are analysed; eight cross-dataset-stable features are selected. Bayesian hyperparameter optimisation with 5-fold stratified CV is applied. Tested across a custom dataset, GramBeddings and PhiUSIIL.

**Advantages:** Directly addresses dataset shift and generalisation; evaluates three independent datasets; XGBoost is efficient for tabular URL features and provides feature importance; reports 90.7% accuracy on the custom test set and 77.8% average accuracy across three independent test sets; substantially faster training than the compared neural network.

**Disadvantages / Limitations:** Cross-dataset accuracy is much lower than within-dataset accuracy, demonstrating that generalisation remains difficult. Average precision and specificity leave room for reducing false positives. URL-centric features may miss webpage HTML, visual and contextual information.

### 3. Comparative Literature Survey

| Paper                                                                  | Algorithm / Methodology                                                                                                                                                                                                                                                                        | Advantages                                                                                                                                                                                           | Disadvantages / Limitations                                                                                                                                                                                                                                            |
|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Phishing Detection System Through Hybrid Machine Learning Based on URL | URL-based supervised ML. Evaluates DT, Logistic Regression, Random Forest, Naive Bayes, Gradient Boosting, KNN and SVC. Proposes the LSD ensemble (LR + SVC + DT) with soft and hard voting. Uses Canopy Feature Selection, Grid Search hyperparameter optimisation and cross-fold validation. | Combines complementary classifiers; systematic feature selection and tuning; compares many baselines; reports accuracy, precision, recall, F1 and specificity; suitable for structured URL features. | Primarily URL-feature based, so it does not directly model rich webpage/visual context. Evaluation is centered on the selected dataset rather than a strong independent cross-dataset test. The ensemble and tuning pipeline is more complex than a single classifier. |

| Paper                                                                                                                   | Algorithm / Methodology                                                                                                                                                                                                                                                                                                                                                                                    | Advantages                                                                                                                                                                                                                                                                                                                                             | Disadvantages / Limitations                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Enhancing Phishing Detection: A Novel Hybrid Deep Learning Framework for Cybercrime Forensics                           | Hybrid deep learning. SMOTE is used for class imbalance. An Ensemble AutoEncoder with ResNet (EARN) performs feature extraction, while a ResNeXt-embedded GRU (RNT) model performs classification. Hyperparameters are optimised using the Jaya optimisation method (RNT-J).                                                                                                                               | Learns nonlinear representations; combines feature extraction and sequence modelling; handles imbalance with SMOTE; includes hyperparameter optimisation; reports 98% accuracy with SMOTE and improved results over comparison models.                                                                                                                 | Architecture is considerably more complex and computationally demanding than conventional ML. Reported results are tied to the experimental dataset, so independent cross-dataset generalisation remains an open concern. Synthetic SMOTE samples may not perfectly reflect real-world phishing distributions. |
| A Study on Adversarial Sample Resistance and Defense Mechanism for Multimodal Learning-Based Phishing Website Detection | Evaluates ML, DL, ensemble and multimodal detectors. Proposes Adversarial Website Generation (AWG), including WGAN-based adversarial sample generation and transfer-based black-box attacks, and studies defense mechanisms. Uses diverse phishing/legitimate sources including OpenPhish, PhishTank, Phishing Database and Alexa.                                                                         | Directly addresses adversarial robustness; tests multiple model families; uses diverse data sources; demonstrates that multimodal detection can be more resistant to adversarial manipulation; provides a practical security perspective beyond ordinary accuracy.                                                                                     | Real adversarial phishing websites are scarce, motivating generated adversarial samples. GAN/WGAN generation and multimodal experiments add substantial complexity and computation. The work targets adversarial robustness and is therefore not a complete substitute for ordinary URL-only detection.        |
| Machine Learning for Early Detection of Phishing URLs in Parked Domains: An Approach Applied to a Financial Institution | Proactive ML framework, "Phishing Hunter", for newly registered and parked domains. Uses preprocessing, feature engineering, feature selection and model optimisation. A Light Gradient Boosting Machine (LightGBM) is the main high-performing classifier. Dataset: 211,659 URLs from real-time SSL certificate monitoring, popular-domain lists and phishing incident reports; 10-fold cross-validation. | Focuses on early detection before widespread blacklisting; large real-world-oriented dataset; combines feature engineering and selection; LightGBM achieved 97.28% accuracy and 96.02% recall on the balanced dataset.                                                                                                                                 | Scope is specifically centred on newly registered/parked domains and a financial-institution use case. Cross-dataset generalisation is not the primary evaluation design. Performance depends on timely availability and quality of domain/SSL-related features.                                               |
| XGBoost-Based URL Phishing Detection Method With Cross-Dataset Validation                                               | X-PHIDE uses XGBoost with allowlist/blocklist checks and URL feature extraction. More than 90 URL/DNS/security/reputation/behaviour features are analysed; eight cross-dataset-stable features are selected. Bayesian hyperparameter optimisation with 5-fold stratified CV is applied. Tested across a custom dataset, GramBeddings and PhiUSIIL.                                                         | Directly addresses dataset shift and generalisation; evaluates three independent datasets; XGBoost is efficient for tabular URL features and provides feature importance; reports 90.7% accuracy on the custom test set and 77.8% average accuracy across three independent test sets; substantially faster training than the compared neural network. | Cross-dataset accuracy is much lower than within-dataset accuracy, demonstrating that generalisation remains difficult. Average precision and specificity leave room for reducing false positives. URL-centric features may miss webpage HTML, visual and contextual information.                              |

## 4. Research Gap Identified

- Dataset dependency: high within-dataset accuracy may not transfer to independent datasets.
- Generalisation: phishing URLs change continuously and previously unseen patterns remain difficult.
- Adversarial vulnerability: attackers can intentionally modify malicious websites to evade AI models.
- Real-time deployment: complex deep-learning pipelines can increase computational cost.
- Explainability: practical users need understandable reasons behind a high-risk prediction.
- Multimodal opportunity: URL-only models may miss HTML, visual and other webpage context.

## 5. References

IEEE Access 11, 36805–36822. DOI: 10.1109/ACCESS.2023.3252366.

IEEE Access 12, 8373–8389. DOI: 10.1109/ACCESS.2024.3351946.

IEEE Access 12, 137805–137824. DOI: 10.1109/ACCESS.2024.3436812.

IEEE Access 13, 145736–145753. DOI: 10.1109/ACCESS.2025.3599454.

IEEE Access 14, 43200–43212. DOI: 10.1109/ACCESS.2026.3672690.

Web sources used for verification: IEEE Xplore records for the five selected papers. The analysis above distinguishes reported methodology/results from limitations inferred from the scope and evaluation design.